Linear Algebra: Homework 4

Yifan Chen

2026 Fall

Problem 1 [6 pts]

Label the following statements as true or false. Assume that V and W are finite-dimensional vector spaces with ordered bases β and γ, respectively, and $T, U : V \to W$ are linear transformations.

(a) For any scalar a, $aT + U$ is a linear transformation from V to W.

(b) $[T]_\beta^\gamma = [U]_\beta^\gamma$ implies that $T = U$.

(c) If $m = \dim(V)$ and $n = \dim(W)$, then $[T]_\beta^\gamma$ is an $m \times n$ matrix.

(d) $[T + U]_\beta^\gamma = [T]_\beta^\gamma + [U]_\beta^\gamma$.

(e) $\mathcal{L}(V, W)$ is a vector space.

(f) $\mathcal{L}(V, W) = \mathcal{L}(W, V)$.

Answer:

The answers are, in order, T, T, F, T, T, F.

Problem 2 [12 pts]

Let β and γ be the standard ordered bases for $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. For each linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$, compute $[T]_\beta^\gamma$.

(a) $T : \mathbb{R}^2 \to \mathbb{R}^3$ defined by $T(a_1, a_2) = (2a_1 - a_2,\ 3a_1 + 4a_2,\ a_1)$.

(b) $T : \mathbb{R}^3 \to \mathbb{R}^2$ defined by $T(a_1, a_2, a_3) = (2a_1 + 3a_2 - a_3,\ a_1 + a_3)$.

(c) $T : \mathbb{R}^3 \to \mathbb{R}$ defined by $T(a_1, a_2, a_3) = 2a_1 + a_2 - 3a_3$.

Answer:

(a)

$$[T]_\beta^\gamma = \begin{pmatrix} 2 & -1 \\ 3 & 4 \\ 1 & 0 \end{pmatrix}.$$

(b)

$$[T]_\beta^\gamma = \begin{pmatrix} 2 & 3 & -1 \\ 1 & 0 & 1 \end{pmatrix}.$$

(c)

$$[T]_\beta^\gamma = \begin{pmatrix} 2 & 1 & -3 \end{pmatrix}.$$

Problem 3 [10 pts]

Define $T : M_{2\times 2}(\mathbb{R}) \to P_2(\mathbb{R})$ by

$$T\begin{pmatrix} a & b \\ c & d \end{pmatrix} = (a+b) + (2d)x + bx^2.$$

Let

$$\beta = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}, \qquad \gamma = \{1, x, x^2\}.$$

Compute $[T]_\beta^\gamma$.

Answer:

$$[T]_\beta^\gamma = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

Problem 4 [10 pts]

Let

$$\alpha = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}, \qquad \beta = \{1, x, x^2\}.$$

Define $T : P_2(\mathbb{R}) \to M_{2\times 2}(\mathbb{R})$ by

$$T(f(x)) = \begin{pmatrix} f'(0) & 2f(1) \\ 0 & f''(3) \end{pmatrix},$$

where $'$ denotes differentiation. Compute $[T]_\beta^\alpha$.

Answer:

$$[T]_\beta^\alpha = \begin{pmatrix} 0 & 1 & 0 \\ 2 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Problem 5 [10 pts]

Let V be an n-dimensional vector space with an ordered basis β. Define $T : V \to \mathbb{F}^n$ by $T(x) = [x]_\beta$. Prove that T is linear.

Answer:

For any $x, y \in V$ and $c \in \mathbb{F}$, let $x = \sum_{i=1}^n x_i\beta_i, y = \sum_{i=1}^n y_i\beta_i$. Then we have $T(cx + y) = [cx + y]_\beta = c[x]_\beta + [y]_\beta = cT(x) + T(y)$, where $[cx + y]_\beta$ and $c[x]_\beta + [y]_\beta$ are both $cx_i + y_i$ in the i-th coordinate.

Therefore, T is linear.

Problem 6 [10 pts]

Let V be an n-dimensional vector space, and let $T : V \to V$ be a linear transformation. Suppose that W is a T-invariant subspace of V (see the exercises of Section 2.1) having dimension k. Show that there is a basis β for V such that $[T]_\beta$ has the form

$$\begin{pmatrix} A & B \\ O & C \end{pmatrix},$$

where A is a $k \times k$ matrix and O is the $(n - k) \times k$ zero matrix.

Answer:

Let $\beta_0 = \{v_1, \ldots, v_k\}$ be a basis of W, and extend it to $\beta = \{v_1, \ldots, v_n\}$ which is a basis of V. For $i \in [1, k]$, we have $v_i \in W$ thus $T(v_i) \in W$ since W is T-invariant, which implies the first k columns only have first k rows non-zero, so $[T]_\beta$ has the required form.

Problem 7 [10 pts]

Let V and W be vector spaces, and let T and U be nonzero linear transformations from V into W. If $\mathcal{R}(T) \cap \mathcal{R}(U) = \{0\}$, prove that $\{T, U\}$ is a linearly independent subset of $\mathcal{L}(V, W)$.

Answer:

Assume $T = cU$ for some $c \in \mathbb{F}$, then $\mathcal{R}(T) = \{cv : v \in \mathcal{R}(U)\}$, since $\mathcal{R}(U)$ is a subspace of W, we have $cv \in \mathcal{R}(U)$ thus $\mathcal{R}(T) \subseteq \mathcal{R}(U)$. Since T is nonzero, we have $\mathcal{R}(T) \cap \mathcal{R}(U) = \mathcal{R}(T) \neq \{0\}$.

Therefore, if $\mathcal{R}(T) \cap \mathcal{R}(U) = \{0\}$, $\{T, U\}$ is linearly independent.

Problem 8 [15 pts]

Let V and W be vector spaces, and let S be a subset of V. Define

$$S^0 = \{T \in \mathcal{L}(V, W) : T(x) = 0 \text{ for all } x \in S\}.$$

Prove the following statements.

(a) S^0 is a subspace of $\mathcal{L}(V, W)$.

Problem 8 [15 pts] continued on next page…

(b) If S_1 and S_2 are subsets of V and $S_1 \subseteq S_2$, then $S_2^0 \subseteq S_1^0$.

(c) If V_1 and V_2 are subspaces of V, then $(V_1 + V_2)^0 = V_1^0 \cap V_2^0$.

Answer:

(a)
- 0 transformation is in S^0.
- For any $T, U \in S^0$ and $x \in S$, $(T+U)(x) = T(x) + U(x) = 0$ thus $T + U \in S^0$.
- For any $T \in S^0$, $c \in \mathbb{F}$ and $x \in S$, $(cT)(x) = cT(x) = 0$ thus $cT \in S^0$.

(b) For any $T \in S_2^0$ and $x \in S_1 \subseteq S_2$, we have $T(x) = 0$ thus $T \in S_1^0$, which implies $S_2^0 \subseteq S_1^0$.

(c) We have $V_1, V_2 \subseteq V_1 + V_2$ since V_1, V_2 are subspaces and $0 \in V_1, V_2$. Therefore, we have $(V_1 + V_2)^0 \subseteq V_1^0 \cap V_2^0$.

For any $T \in V_1^0 \cap V_2^0$ and $x \in V_1 + V_2$, let $x = x_1 + x_2$ where $x_1 \in V_1$ and $x_2 \in V_2$, we have $T(x) = T(x_1) + T(x_2) = 0$ since T is linear, thus $T \in (V_1 + V_2)^0$. Therefore, we have $V_1^0 \cap V_2^0 \subseteq (V_1 + V_2)^0$.

In conclusion, we have $(V_1 + V_2)^0 = V_1^0 \cap V_2^0$.

Problem 9 [10 pts]

Let V and W be vector spaces such that $\dim(V) = \dim(W)$, and let $T : V \to W$ be linear. Show that there exist ordered bases β and γ for V and W, respectively, such that $[T]_\beta^\gamma$ is a diagonal matrix.

Answer:

Let $k = \text{rank}(T)$, arbitrarily choose a basis $\beta_0 = \{u_1, \ldots, u_n\}$ of V and choose k linearly independent elements of $T(\beta_0)$ as basis $\gamma_0 = \{v_1, \ldots, v_k\}$ of $\mathcal{R}(T)$. Since $\mathcal{R}(T)$ is a subspace of W, we can extend γ_0 into $\gamma = \{v_1, \ldots, v_n\}$ which is a basis of W.

We rearrange β_0 into $\beta = \{u_1', \ldots, u_n'\}$:

- Rearrange β_0, put the k chosen elements at the beginning so that $T(u_i') = v_i$ for $i \in [1, k]$.
- For $i \in (k, n]$, assume $T(u_i') = \sum_{j=1}^k a_{i,j} v_j$, then let $u_i' \leftarrow u_i' - \sum_{j=1}^k a_{i,j} u_j'$, then we have $T(u_i') = 0$ by linearity of T.

β is a basis: every u_i can be written as a linear combination of β, so β spans V; if β is linearly dependent, then β_0 is linearly dependent as well.

Therefore, $[T]_\beta^\gamma = \text{diag}(1, \ldots, 1, 0, \ldots, 0)$ with k ones.

Problem 10 [10 pts]

Label the following statements as true or false. In each part, V, W, and Z denote vector spaces with ordered (finite) bases α, β, and γ, respectively; $T : V \to W$ and $U : W \to Z$ denote linear transformations; and A and B denote matrices.

(a) $[UT]_\alpha^\gamma = [T]_\alpha^\beta [U]_\beta^\gamma$.

Problem 10 [10 pts] continued on next page…

(b) $[T(v)]_\beta = [T]_\alpha^\beta [v]_\alpha$ for all $v \in V$.

(c) $[U(w)]_\beta = [U]_\alpha^\beta [w]_\beta$ for all $w \in W$.

(d) $[I_V]_\alpha = I$.

(e) $[T^2]_\alpha^\beta = \left([T]_\alpha^\beta\right)^2$.

(f) $A^2 = I$ implies that $A = I$ or $A = -I$.

(g) $T = L_A$ for some matrix A.

(h) $A^2 = O$ implies that $A = O$, where O denotes the zero matrix.

(i) $L_{A+B} = L_A + L_B$.

(j) If A is square and $A_{ij} = \delta_{ij}$ for all i and j, then $A = I$.

Answer:

The answers are, in order, (a. to e.) F, T, F, T, F, (f. to j.) F, F, F, T, T.

Problem 11 [16 pts]

(a) Let

$$A = \begin{pmatrix} 1 & 3 \\ 2 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 & -3 \\ 4 & 1 & 2 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 1 & 4 \\ -1 & -2 & 0 \end{pmatrix}, \qquad D = \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}.$$

Compute $A(2B + 3C)$, $(AB)D$, and $A(BD)$.

(b) Let

$$A = \begin{pmatrix} 2 & 5 \\ -3 & 1 \\ 4 & 2 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & -2 & 0 \\ 1 & -1 & 4 \\ 5 & 5 & 3 \end{pmatrix}, \qquad C = \begin{pmatrix} 4 & 0 & 3 \end{pmatrix}.$$

Compute $A^\top$, $A^\top B$, $BC^\top$, CB, and CA.

Answer:

(a)

$$A(2B + 3C) = \begin{pmatrix} 1 & 3 \\ 2 & -1 \end{pmatrix} \cdot \begin{pmatrix} 5 & 3 & 6 \\ 5 & -4 & 4 \end{pmatrix} = \begin{pmatrix} 20 & -9 & 18 \\ 5 & 10 & 8 \end{pmatrix},$$

$$(AB)D = \begin{pmatrix} 13 & 3 & 3 \\ -2 & -1 & -8 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix} = \begin{pmatrix} 29 \\ -26 \end{pmatrix},$$

$$BD = \begin{pmatrix} 1 & 0 & -3 \\ 4 & 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix} = \begin{pmatrix} -7 \\ 12 \end{pmatrix},$$

$$A(BD) = \begin{pmatrix} 1 & 3 \\ 2 & -1 \end{pmatrix} \cdot \begin{pmatrix} -7 \\ 12 \end{pmatrix} = \begin{pmatrix} 29 \\ -26 \end{pmatrix}.$$

Problem 11 [16 pts] continued on next page…

(b)

$$A^\top = \begin{pmatrix} 2 & -3 & 4 \\ 5 & 1 & 2 \end{pmatrix},$$

$$A^\top B = \begin{pmatrix} 2 & -3 & 4 \\ 5 & 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} 3 & -2 & 0 \\ 1 & -1 & 4 \\ 5 & 5 & 3 \end{pmatrix} = \begin{pmatrix} 23 & 19 & 0 \\ 26 & -1 & 10 \end{pmatrix},$$

$$BC^\top = \begin{pmatrix} 3 & -2 & 0 \\ 1 & -1 & 4 \\ 5 & 5 & 3 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 12 \\ 16 \\ 29 \end{pmatrix},$$

$$CB = \begin{pmatrix} 4 & 0 & 3 \end{pmatrix} \cdot \begin{pmatrix} 3 & -2 & 0 \\ 1 & -1 & 4 \\ 5 & 5 & 3 \end{pmatrix} = \begin{pmatrix} 27 & 7 & 9 \end{pmatrix},$$

$$CA = \begin{pmatrix} 4 & 0 & 3 \end{pmatrix} \cdot \begin{pmatrix} 2 & 5 \\ -3 & 1 \\ 4 & 2 \end{pmatrix} = \begin{pmatrix} 20 & 26 \end{pmatrix}.$$

Problem 12 [20 pts]

Let $g(x) = 3 + x$. Let $T : P_2(\mathbb{R}) \to P_2(\mathbb{R})$ and $U : P_2(\mathbb{R}) \to \mathbb{R}^3$ be the linear transformations respectively defined by

$$T(f(x)) = f'(x)g(x) + 2f(x) \qquad \text{and} \qquad U(a + bx + cx^2) = (a + b, c, a - b).$$

Let β and γ be the standard ordered bases of $P_2(\mathbb{R})$ and $\mathbb{R}^3$, respectively.

(a) Compute $[U]_\beta^\gamma$, $[T]_\beta$, and $[UT]_\beta^\gamma$ directly. Then use Theorem 2.11 to verify your result.

(b) Let $h(x) = 3 - 2x + x^2$. Compute $[h(x)]_\beta$ and $[U(h(x))]_\gamma$. Then use $[U]_\beta^\gamma$ from (a) and Theorem 2.14 to verify your result.

Answer:

(a) We have $UT(a + bx + cx^2) = U((2a + 3b) + (3b + 6c)x + 4cx^2) = (2a + 6b + 6c, 4c, 2a - 6c)$, thus

$$[U]_\beta^\gamma = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix},$$

$$[T]_\beta = \begin{pmatrix} 2 & 3 & 0 \\ 0 & 3 & 6 \\ 0 & 0 & 4 \end{pmatrix},$$

$$[UT]_\beta^\gamma = \begin{pmatrix} 2 & 6 & 6 \\ 0 & 0 & 4 \\ 2 & 0 & -6 \end{pmatrix},$$

Problem 12 [20 pts] continued on next page…

$$[U]_\beta^\gamma[T]_\beta = \begin{pmatrix}1 & 1 & 0\\ 0 & 0 & 1\\ 1 & -1 & 0\end{pmatrix}\cdot\begin{pmatrix}2 & 3 & 0\\ 0 & 3 & 6\\ 0 & 0 & 4\end{pmatrix} = \begin{pmatrix}2 & 6 & 6\\ 0 & 0 & 4\\ 2 & 0 & -6\end{pmatrix} = [UT]_\beta^\gamma,$$

which verifies the theorem.

(b) We have $U(h(x)) = U(3-2x+x^2) = (1,1,5)$, thus

$$[h(x)]_\beta = \begin{pmatrix}3\\ -2\\ 1\end{pmatrix},$$

$$[U(h(x))]_\gamma = \begin{pmatrix}1\\ 1\\ 5\end{pmatrix},$$

$$[U]_\beta^\gamma[h(x)]_\beta = \begin{pmatrix}1 & 1 & 0\\ 0 & 0 & 1\\ 1 & -1 & 0\end{pmatrix}\cdot\begin{pmatrix}3\\ -2\\ 1\end{pmatrix} = \begin{pmatrix}1\\ 1\\ 5\end{pmatrix} = [U(h(x))]_\gamma,$$

which verifies the theorem.

Problem 13 [20 pts]

Let α and β be the ordered bases given in Problem 4. For each of the following parts, use Theorem 2.14 to compute the indicated vector for the linear transformation T defined below.

(a) $[T(A)]_\alpha$, where

$$A = \begin{pmatrix}1 & 4\\ -1 & 6\end{pmatrix}.$$

Here $T : M_{2\times2}(\mathbb{F}) \to M_{2\times2}(\mathbb{F})$ is defined by $T(A) = A^\top$.

(b) $[T(f(x))]_\alpha$, where $f(x) = 4 - 6x + 3x^2$. Here $T : P_2(\mathbb{R}) \to M_{2\times2}(\mathbb{R})$ is defined by

$$T(f(x)) = \begin{pmatrix}f'(0) & 2f(1)\\ 0 & f''(3)\end{pmatrix}.$$

Answer:

(a)

$$[T(A)]_\alpha = [T]_\alpha[A]_\alpha = \begin{pmatrix}1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1\end{pmatrix}\cdot\begin{pmatrix}1\\ 4\\ -1\\ 6\end{pmatrix} = \begin{pmatrix}1\\ -1\\ 4\\ 6\end{pmatrix}.$$

Problem 13 [20 pts] continued on next page…

(b)

$$[T(f(x))]_\alpha = [T]^\alpha_\beta [f(x)]_\beta = \begin{pmatrix} 0 & 1 & 0 \\ 2 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ -6 \\ 3 \end{pmatrix} = \begin{pmatrix} -6 \\ 2 \\ 0 \\ 6 \end{pmatrix}.$$

Problem 14 [10 pts]

Find linear transformations $U, T : \mathbb{F}^2 \to \mathbb{F}^2$ such that $UT = T_0$ (the zero transformation) but $TU \neq T_0$. Use your answer to find matrices A and B such that $AB = O$ but $BA \neq O$.

Answer:

Consider $T(a, b) = (a, 0)$ and $U(a, b) = (b, 0)$; we have $UT(a, b) = U(a, 0) = (0, 0)$ and $TU(a, b) = T(b, 0) = (b, 0)$. Therefore, $UT = T_0$ and $TU \neq T_0$.

Let β be the standard basis of $\mathbb{F}^2$, then let

$$A = [U]_\beta = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

$$B = [T]_\beta = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

We have $AB = O$ but $BA = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq O$.

Problem 15 [15 pts]

Let V, W, and Z be vector spaces, and let $T : V \to W$ and $U : W \to Z$ be linear.

(a) Prove that if UT is one-to-one, then T is one-to-one. Must U also be one-to-one?

(b) Prove that if UT is onto, then U is onto. Must T also be onto?

(c) Prove that if U and T are one-to-one and onto, then UT is also.

Answer:

(a) If $T(x) = T(y)$, then $UT(x) = U(T(x)) = U(T(y)) = UT(y)$. Since UT is one-to-one, we obtain $x = y$ and thus T is one-to-one. U is not required to be one-to-one since $\mathcal{R}(T)$ may not equal W.

(b) Since UT is onto, for any $z \in Z$, there is some x such that $z = UT(x) = U(T(x))$, then $y = T(x)$ satisfies $U(y) = z$ thus U is onto. T is not required to be onto since $\mathcal{N}(U)$ may not equal $\{0\}$ thus only one element of $\mathcal{N}(U)$ need be in $\mathcal{R}(T)$.

(c) If $UT(x) = UT(y)$, then $T(x) = T(y)$ since U is one-to-one and $x = y$ since T is one-to-one, thus UT is one-to-one. For any $z \in Z$, there is some $y \in W$ such that $U(y) = z$, and some $x \in V$ such that $T(x) = y$, then $UT(x) = U(y) = z$, thus UT is onto.

Problem 16 [15 pts]

Let V be a finite-dimensional vector space, and let $T : V \to V$ be linear.

(a) If $\text{rank}(T) = \text{rank}(T^2)$, prove that $\mathcal{R}(T) \cap \mathcal{N}(T) = \{0\}$. Deduce that $V = \mathcal{R}(T) \oplus \mathcal{N}(T)$ (see the exercises of Section 1.3).

(b) Prove that $V = \mathcal{R}(T^k) \oplus \mathcal{N}(T^k)$ for some positive integer k.

Answer:

(a) Let $W = \mathcal{R}(T)$ and let T_W be the restriction of T to W; then W is T-invariant. $\mathcal{R}(T^2) = \mathcal{R}(T_W T)$ is a subspace of $W = \mathcal{R}(T)$, and combined with $\text{rank}(T) = \text{rank}(T^2)$ this shows that T_W is one-to-one and onto. If $x \in \mathcal{R}(T) \cap \mathcal{N}(T)$, then $x \in \mathcal{N}(T_W)$, thus $x = 0$ since T_W is one-to-one. Therefore, $\mathcal{R}(T) \cap \mathcal{N}(T) = \{0\}$.

For any $x \in V$, since $T(x) \in W$ and T_W is onto, there exists some $y \in W$ such that $T(x) = T_W(y) = T(y)$. Since T is linear, we have $T(x - y) = T(x) - T(y) = 0$ thus $x - y \in \mathcal{N}(T)$. Therefore, $V = \mathcal{R}(T) + \mathcal{N}(T)$.

Combining the above arguments, we obtain $V = \mathcal{R}(T) \oplus \mathcal{N}(T)$.

(b) We have $\text{rank}(T^{k+1}) \leq \text{rank}(T^k)$ since $\mathcal{R}(T^{k+1})$ is a subspace of $\mathcal{R}(T^k)$. Therefore, there must be some k such that $\text{rank}(T^{k+1}) = \text{rank}(T^k)$. Let $W = \mathcal{R}(T^k)$ and apply the same argument as in part (a); we obtain the same result $V = \mathcal{R}(T^k) \oplus \mathcal{N}(T^k)$.

Problem 17 [9 pts]

Label the following statements as true or false. In each part, V and W are vector spaces with ordered (finite) bases α and β, respectively, $T : V \to W$ is linear, and A and B are matrices.

(a) $\left([T]_\alpha^\beta\right)^{-1} = [T^{-1}]_\alpha^\beta$.

(b) T is invertible if and only if T is one-to-one and onto.

(c) $T = L_A$, where $A = [T]_\alpha^\beta$.

(d) $M_{2\times 3}(\mathbb{F})$ is isomorphic to $\mathbb{F}^5$.

(e) $P_n(\mathbb{F})$ is isomorphic to $P_m(\mathbb{F})$ if and only if $n = m$.

(f) $AB = I$ implies that A and B are invertible.

(g) If A is invertible, then $(A^{-1})^{-1} = A$.

(h) A is invertible if and only if L_A is invertible.

(i) A must be square in order to possess an inverse.

Answer:

The answers are, in order, (a. to e.) F, T, F, F, T, (f. to i.) F, T, T, T.

Problem 18 [15 pts]

For each of the following linear transformations T, determine whether T is invertible and justify your answer.

(a) $T : \mathbb{R}^2 \to \mathbb{R}^3$ defined by $T(a_1, a_2) = (a_1 - 2a_2,\ a_2,\ 3a_1 + 4a_2)$.

(b) $T : \mathbb{R}^3 \to \mathbb{R}^3$ defined by $T(a_1, a_2, a_3) = (3a_1 - 2a_3,\ a_2,\ 3a_1 + 4a_2)$.

(c) $T : M_{2\times 2}(\mathbb{R}) \to P_2(\mathbb{R})$ defined by

$$T\begin{pmatrix} a & b \\ c & d \end{pmatrix} = a + 2bx + (c + d)x^2.$$

Answer:

(a) T is not invertible since $\dim \mathbb{R}^2 \neq \dim \mathbb{R}^3$.

(b) T is invertible. We have $\mathcal{N}(T) = \{0\}$ and $\dim \mathbb{R}^3 = \dim \mathbb{R}^3$, thus T is one-to-one and onto.

(c) T is not invertible since $\dim M_{2\times 2}(\mathbb{R}) \neq \dim P_2(\mathbb{R})$.

Problem 19 [5 pts]

Let A be invertible. Prove that $A^\top$ is invertible and $(A^\top)^{-1} = (A^{-1})^\top$.

Answer:

Let $B = A^{-1}$, then we have $AB = BA = I$. Take the transpose and we have $B^\top A^\top = A^\top B^\top = I^\top = I$. Therefore, we have $(A^\top)^{-1} = B^\top = (A^{-1})^\top$.

Problem 20 [10 pts]

Let A and B be $n \times n$ matrices such that AB is invertible. Prove that A and B are invertible. Give an example to show that arbitrary matrices A and B need not be invertible if AB is invertible.

Answer:

Since AB is invertible, L_{AB} is one-to-one and onto. By Problem 15, L_A is onto and L_B is one-to-one. Therefore, A and B are both invertible.

Consider $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$; then $AB = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is invertible, while A, B are not since they are not square.

Problem 21 [10 pts]

For an arbitrary $n \times n$ matrix $A \in M_{n\times n}(\mathbb{R})$ with all entries smaller than 1 for their absolute values, can we have

$$\lim_{m\to\infty} A^m \to 0?$$

Justify your answer.

Answer:

No. Assume all the entries of A equal $a_1 = v \in (0, 1)$; then all the entries of A^2 equal $a_2 = na_1^2 = nv^2$, and as m increases the entries satisfy $a_{m+1} = nva_m$, so $a_m = (nv)^m/n$. For $n > v^{-1}$ we have $nv > 1$, hence the entries diverge to infinity.

Problem 22 [10 pts]

Let U be a subspace of a finite-dimensional vector space V. Let $T : V \to V$ be a linear transformation. Prove

$$\dim(U) - \dim\big(\mathcal{N}(T)\big) \le \dim\big(T(U)\big) \le \dim(U).$$

Answer:

Let $T_U : U \to V$ be the restriction of T to U. Then we obtain $\dim(T(U)) = \text{rank}(T_U) \le \dim(U)$ and $\text{nullity}(T_U) \le \text{nullity}(T) = \dim(\mathcal{N}(T))$, thus $\dim(T(U)) + \dim(\mathcal{N}(T)) \ge \text{rank}(T_U) + \text{nullity}(T_U) = \dim(U)$.

Therefore, we have $\dim(U) - \dim(\mathcal{N}(T)) \le \dim(T(U)) \le \dim(U)$.

Problem 23 [10 pts]

Prove

$$\text{rank}(L_A) + \text{rank}(L_B) - n \le \text{rank}(L_{AB}) \le \min\{\text{rank}(L_A), \text{rank}(L_B)\},$$

where A and B are $n \times n$ matrices.

Answer:

Let $U = \mathcal{R}(L_B)$; we have $\mathcal{R}(L_{AB}) = L_A(U) \subseteq \mathcal{R}(L_A)$, thus $\dim(U) - \dim(\mathcal{N}(L_A)) \le \dim(L_A(U)) \le \dim(U)$, which is equivalent to $\text{rank}(L_B) - (\dim(L_A) - \text{rank}(L_A)) \le \text{rank}(L_{AB}) \le \text{rank}(L_B)$.

Combine with $\mathcal{R}(L_{AB}) \subseteq \mathcal{R}(L_A)$, which implies $\text{rank}(L_{AB}) \le \text{rank}(L_A)$, we have $\text{rank}(L_A) + \text{rank}(L_B) - n \le \text{rank}(L_{AB}) \le \min\{\text{rank}(L_A), \text{rank}(L_B)\}$.

Problem 24 [10 pts]

Prove

$$\text{rank}(L_{A+B}) \leq \text{rank}(L_A) + \text{rank}(L_B)$$

where A and B are $n \times m$ matrices.

Answer:

Let $\alpha = \{u_1, \ldots, u_k\}$ be a basis of $\mathcal{R}(L_A)$ and $\beta = \{v_1, \ldots, v_l\}$ be a basis of $\mathcal{R}(L_B)$. For any $x \in \mathbb{F}^m$, $L_{A+B}(x) = (A+B)x = Ax + Bx$ and can be expressed as a linear combination of $\alpha \cup \beta$, thus $\mathcal{R}(L_{A+B}) \subseteq \text{Span}(\alpha \cup \beta)$.

Therefore, $\text{rank}(L_{A+B}) \leq k + l = \text{rank}(L_A) + \text{rank}(L_B)$.